

How intellectual property protection enhances data capitalization: The mediating effects of resources and willingness to capitalize on data

Yu Wang^a, Fan Dong^{b,*}

^a School of Economics, Shenzhen Polytechnic University, Shenzhen, China

^b School of Law and Intellectual Property, Guangdong Polytechnic Normal University, Guangzhou, China

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ABSTRACT

While intellectual property (IP) protection provides legal mechanisms to safeguard assets, the majority of previous research has focused on its benefits for traditional assets, neglecting its influence on facilitating data capitalization. Using a longitudinal dataset of Chinese listed firms from 2011 to 2022, this study demonstrates that IP protection has a positive effect on corporate data capitalization by enhancing data capitalization resources and willingness. Moreover, heterogeneity analysis reveals that private firms, high-tech industries, and regions with advanced digital infrastructure are more effective in achieving data capitalization under IP protection. Findings contribute to the literature on IP and data capitalization and provide practical insights for practitioners seeking to optimize data-driven strategies.

1. Introduction

Data capitalization have emerged as a critical catalyst for enterprise growth (Plekhanov et al., 2023). Ongoing digital transformation has elevated the role of data beyond merely supplementing conventional innovation inputs, actively reconfiguring innovation paradigms and enhancing organizational outcomes (Raguseo, 2018). Forward-thinking firms increasingly consider digital capability development as essential for sustaining competitiveness. This competitive advantage is attributable not simply to adopting cutting-edge technologies, such as artificial intelligence (AI) and big data analytics, but more fundamentally to establishing sophisticated frameworks for data acquisition and governance. As such, organizations must adopt systematic approaches to convert digital resources into valuable assets (Hu et al., 2022).

Previous studies have conceptualized this transformation as involving the structured documentation and purposeful organization of digital infrastructure, including software systems and electronic records, to establish sustainable, long-term competitive platforms that support strategic decision-making and innovation (Birch et al., 2021; Hu et al., 2022). Empirical research has identified several key benefits arising from effective data capitalization, such as strengthening professional networks through improved data sharing and collaboration (Kozjek et al., 2018), enhancing firms' innovation capabilities by enabling better use of data-driven insights (Lyytinen et al., 2016), and improving

financial resilience amid economic downturns by enabling more adaptive and informed responses (Brynjolfsson & McElheran, 2016).

While the strategic value of data capitalization has been widely acknowledged in academic and managerial discourse, many organizations continue to encounter substantial barriers when attempting to implement such practices effectively (Boudreau et al., 2022). A key challenge is the unique nature of digital information as, unlike physical assets, it is highly replicable, easily transferable, and can be disseminated at virtually no cost. While enabling scalability and innovation, these characteristics also expose organizations to heightened risks of unauthorized access, data leakage, and competitors' imitation. This inherent vulnerability necessitates a shift in research focus from merely identifying and promoting the benefits of data capitalization to constructing more comprehensive and actionable frameworks to address associated risks, particularly in terms of data protection, governance, and security enhancement.

In this context, intellectual property (IP) regimes are critical institutional safeguards that offer legal protection for digital assets. The growing centrality of data-driven innovation in contemporary business operations has subsequently intensified scholarly interest in exploring the multifaceted influence of IP protection on facilitating technological advancement (Ding & Yang, 2025; Hu et al., 2022; Miric et al., 2019; Wen & Deng, 2023; Zheng et al., 2023). IP protection is a foundational pillar of corporate innovation and long-term development. While IP

* Corresponding author.

E-mail address: dongfan@gpnu.edu.cn (F. Dong).

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systems undeniably contribute to organizational innovation ecosystems, their application to digital assets introduces distinct challenges. Fundamental differences between data resources and traditional physical assets have established theoretical and practical gaps in our understanding of how IP protection influences data management and utilization. Previous literature remains inconclusive regarding the exact nature of this relationship, giving rise to our core research question: *How does IP protection affect data capitalization in the digital economy?*

To address this question, we begin by developing hypotheses grounded in theoretical analysis in Section 2, focusing on two potential mediating pathways. Section 3 presents our research methodology, and we empirically investigate the relationship between IP protection and corporate data capitalization using data from Chinese listed firms in Section 4, followed by a heterogeneity analysis in Section 4 to explore variations across different contexts. Finally, Section 6 examines the significance of our empirical findings, contributing to the expanding body of literature on data capitalization and IP protection.

2. Literature review and hypotheses development

2.1. IP protection

The institutional framework of IP rights is a fundamental pillar supporting innovation ecosystems and sustainable economic progress. This consensus within scholarly discourse has strong theoretical grounding in endogenous growth models, particularly [Romer's \(1986\)](#) seminal work that emphasized technology-driven development pathways. Contemporary research further determined that robust IP regimes have dual critical functions of shielding proprietary knowledge assets while also stimulating corporate innovation investments ([Wen & Deng, 2023](#)). Empirical evidence has consistently demonstrated IP systems' effectiveness in preserving technological innovations ([Gaessler et al., 2021](#)) and accelerating innovation cycles ([Lyytinen et al., 2016](#); [Song et al., 2024](#)).

The dynamic evolution of IP frameworks reflects ongoing efforts to address technological disruptions. The digital transformation era has witnessed significant global regulatory innovations, which are exemplified by the European Union's General Data Protection Regulation and China's 2017 Data Security Law, both of which established critical governance structures for digital assets. However, academic inquiry has not sufficiently addressed the distinctive challenges posed by digital economy conditions. In contrast to conventional innovation contexts, digital firms confront increased risks from the nonexcludable characteristics of data resources ([Brynjolfsson & McElheran, 2016](#)) that dramatically lower replication barriers. This research gap underscores the imperative of systematically investigating how IP mechanisms function within data-driven economic paradigms.

Contemporary scholarship has increasingly focused on the transformative functions of IP frameworks in the digital era. Current empirical studies have demonstrated that effective IP regimes can bridge information gaps between corporations and capital markets and ensure data security amid intensifying market competition ([Wang, 2021](#)). Particularly noteworthy is the application of IP principles to intangible digital assets. For example, previous research has revealed that algorithmic protections not only improve AI-related patents' commercialization rates but also establish substantial barriers against imitation ([Gambardella, 2023](#)). Nevertheless, the academic discourse has offered a significant counter-narrative. Critical examinations by [Bessen and Maskin \(2009\)](#) revealed a potential paradox in which the very protections designed to stimulate innovation may inadvertently constrain technological advancement in nascent fields. The authors' analysis indicated that expansive patent coverage in areas such as machine intelligence could paradoxically suppress innovation by discouraging knowledge sharing and restricting the cross-pollination of ideas essential for breakthrough developments. This dichotomy underscores the need for balanced intellectual property approaches that reconcile protection

with collaboration imperatives.

2.2. Data capitalization

Data capitalization are the process through which firms convert documents, software, platforms, and electronic data into valuable data assets via electronic recording, establishing a foundation for future competitive advantages ([Birch et al., 2021](#); [Hu et al., 2022](#)). Once considered a byproduct in the realms of technological innovation and strategic management ([Brynjolfsson & McElheran, 2016](#)), data capitalization has become a central focus of academic research. In the contemporary competitive landscape, relying on traditional physical assets is no longer sufficient. Instead, data assets have become increasingly essential ([Xia et al., 2023](#)). As a result, the "assetization" of data are critical for firms seeking to maintain a competitive edge ([Hu et al., 2022](#); [Jackson & Luu, 2023](#)). However, this process is far from straightforward and requires firms' sustained investment in digital technology development. [Lyytinen et al. \(2016\)](#) outlined four distinct stages of data capitalization, emphasizing that achieving this transformation is complex and necessitates support from various driving factors.

The driving force behind organizational data capitalization are fundamentally associated with breakthroughs in digital innovation. Cutting-edge technological developments-spanning Internet of Things (IoT) networks, distributed cloud architecture, and advanced machine learning algorithms have revolutionized how firms acquire, manage, and extract insights from data resources ([Constantinides et al., 2018](#)). These technological enablers facilitate more sophisticated investment frameworks and incentivize businesses to cultivate digital asset portfolios. Nevertheless, technological adoption alone is inadequate for successful transformation as firms must develop organizational competencies to effectively embed emerging digital solutions into core business processes. Previous research has underscored the criticality of dynamic capabilities, particularly the capacity to institutionalize data-driven decision-making throughout the organizational hierarchy ([Teece, 2018](#)). This metacapability determines how effectively firms can operationalize their technological investments. Sector-specific considerations further complicate implementation. For example, healthcare organizations must navigate complex ethical dilemmas when monetizing patient data ([Vezyridis & Timmons, 2021](#)). Similarly, energy sector organizations face unique challenges that require holistic strategies to balance technological deployment with ecosystem collaboration and public interest alignment. These contextual factors necessitate tailored approaches to data value realization across industries.

In summary, while existing scholarship has made significant strides in mapping the organizational and technological drivers of data capitalization, the institutional dimension, particularly how regulatory frameworks and IP regimes influence this process, constitutes a notable research gap. This study addresses this underexplored area by analyzing how IP protection influences corporate data capitalization through the lens of institutional economics.

2.3. Hypothesis development

Endogenous growth theory emphasizes that technological progress is a critical driver of economic development ([Romer, 1986](#)). In the context of this study, IP protection has a vital influence on corporate technological research and innovation ([Wen & Deng, 2023](#)), which subsequently enhances corporate innovation activities. Therefore, this study highlights the significance of IP protection in facilitating corporate data capitalization.

First, robust IP protection effectively delays competitors' imitation, enabling firms to establish and sustain competitive advantages. Unlike physical assets, the intangible and fluid nature of digital assets makes them more vulnerable to competitors' dissemination and replication ([Hu et al., 2022](#); [Li & Yuan, 2024](#)). As the digital economy continues to

evolve, theoretical and practical research has increasingly emphasized the significance of IP protection. When enterprises face competitive imitation, strong IP protection can prevent the leakage or unauthorized replication of their data assets. For example, [Trequattrini et al. \(2022\)](#) found that IP protection in the music recording industry significantly reduces the likelihood of competitors imitating technologies, facilitating the industry's digital transformation. In this way, IP protection provides a secure institutional foundation for corporate digital innovation and transformation.

Second, IP protection lowers information disclosure costs and alleviates corporate financing constraints. As IP regulations evolve, the legal distinctions between existing (i.e., physical assets) and new (i.e., digital assets) technologies have become clearer ([Nagaraj, 2018](#)), offering more effective and transparent institutional guarantees for information disclosure. Reduced disclosure costs promote increased interaction between firms and stakeholders, broadening financing channels and supporting data assets' development. [Zheng et al. \(2023\)](#) found that IP protection enhances firms' ability to secure financing, advancing its digital transformation. Therefore, IP protection establishes digital assets by reducing information asymmetry and expanding financing opportunities. Finally, IP protection is an institutional safeguard that ensures inventors receive fair compensation, which promotes digital asset development. For example, in the software industry, the implementation of IP laws has been demonstrated to increase the likelihood that firms will develop proprietary software solutions, accelerating digital innovations' commercialization ([Wen & Deng, 2023](#)). Therefore, firms under effective IP protection are better positioned to generate sustained revenue from their digital assets. Therefore, we propose the hypothesis as follows:

H1. IP protection promotes corporate data capitalization.

We also posit that IP protection promotes corporate data assets through two primary pathways. The first pathway is enhancing corporate data capitalization by facilitating the accumulation and use of data resources, strengthening information security, particularly in terms of data privacy and security, which is crucial for firms to confidently invest in data infrastructure and analytical capabilities ([Hu et al., 2022](#)). These efforts improve data assets' quality and reliability and increase their value in the data capitalization process. Second, IP protection enhances corporate data management capabilities, establishing a foundation for accumulating data resources and facilitating data capitalization. Through formal and informal IP protection, firms can progressively strengthen data management and control capabilities, supporting the smooth progression of data capitalization ([Miric et al., 2019](#)). For instance, when firms develop new data analysis tools or technologies, they can protect these innovations through patents, ensuring market competitiveness. This not only encourages firms to invest in data management but also enhances the potential value of their data assets ([Zheng et al., 2023](#)). Finally, from the perspective of stakeholder interaction, IP protection provides security for corporate cooperation and data sharing. Firms that safeguard against data infringements are better positioned to collaborate with stakeholders and identify new business opportunities ([Hertzfeld et al., 2006](#); [Wall, 2018](#)). Such cooperation is crucial for the effective utilization of data resources and the advancement of data capitalization. Therefore, we propose the hypothesis as follows:

H2. IP protection promotes firms' data capitalization by improving data capitalization resources.

The second pathway is IP protection enhancing corporate data capitalization by increasing firms' willingness to engage in data capitalization. From the perspective of perceived uncertainty, previous studies have shown that IP protection reduces uncertainty regarding data ownership and usage, whether digital or physical ([Ginarte & Park, 1997](#); [Moerchel et al., 2022](#)). When firms are confident that their innovations in data technology and other areas are fully protected by legal

safeguards, they are more likely to increase investment in digital infrastructure and initiatives. Effective IP protection alleviates concerns regarding technology leakage and the theft of innovation outcomes, boosting firms' enthusiasm for capitalizing on innovations ([Hu et al., 2022](#); [Kollock & Dellermann, 2018](#); [Trequattrini et al., 2022](#)). When firms are confident that their data innovations and technologies are safeguarded by a robust IP system, they are more inclined to invest in digital infrastructure and explore new opportunities for data capitalization. Furthermore, from the perspective of competitive advantage, IP protection establishes a relatively secure innovation environment, reducing the risk of external competitors' imitation. This enables firm managers to anticipate future competitive advantages and motivates additional investment in digital assets rather than traditional physical assets ([Lin, 2023](#); [Sampat & Williams, 2019](#)). [Wen and Deng \(2023\)](#) found that strong IP protection can drive corporate investment in research and development (R&D), demonstrating that managers consider IP as an external condition for securing competitive advantages. Therefore, we propose the hypothesis as follows:

H3. IP protection promotes data capitalization by improving firms' data capitalization willingness.

[Fig. 1](#) illustrates our overall theoretical framework.

3. Methodology

3.1. Data sources

We build this study's empirical foundation on a sample of Chinese A-share listed firms from 2011 to 2022 with data collection and processing following established academic protocols. The research team developed a novel data asset index through systematic quantitative textual content analysis of corporate annual reports, using structured data extraction methodologies. Complementary financial and governance variables were obtained from authoritative databases, including Wind and China Stock Market & Accounting Research databases.

Following established practices in corporate finance research ([Geng et al., 2025](#); [Xiong et al., 2024](#); [Xue et al., 2024](#)), our sample excludes financial institutions due to their distinct regulatory environment, in addition to specially designated (ST/*ST) firms indicating financial distress, and those with incomplete records. These exclusions were implemented through comprehensive missing value diagnostics, outlier detection, and data consistency checks. The final dataset includes 28,908 high-quality firm-year observations, offering (1) 12 years of continuous panel data for longitudinal analysis, (2) broad cross-sectoral representation of China's nonfinancial listed firms, and (3) robust control variables to address potential bias. This carefully constructed sample addresses critical econometric concerns related to selection bias, data quality, and industry-specific distortions, while ensuring comparability with existing empirical work in the field.

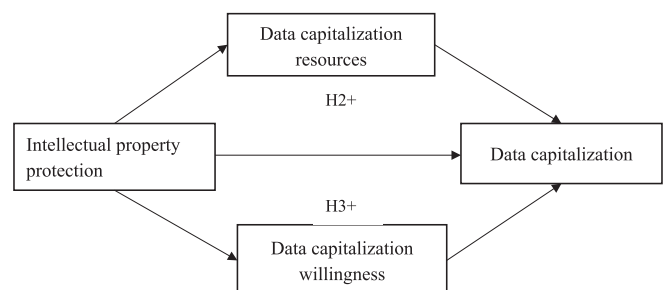


Fig. 1. Conceptual framework.

3.2. Variables

3.2.1. Dependent variable

Referencing previous literature (Hu et al., 2022), we operationalize data capitalization as a multiphase transformation process that includes (1) systematic digital acquisition of operational metrics, where value emerges through cumulative data aggregation; (2) rigorous information processing involving cryptographic encoding and privacy-preserving protocols; (3) advanced analytical transformation using machine learning algorithms; and (4) strategic business integration for optimized decision-making. Our quantification methodology examines financial disclosures to identify and aggregate data-related capital expenditures, calculating their weighted contribution to total fixed and intangible assets. This approach captures the technological infrastructure investments (servers, software) and intellectual capital development (algorithms, analytics) that constitute an organization's data asset portfolio.

When financial disclosures of fixed or intangible asset expenditures contain terminology associated with data capitalization (as specified in Table 1), this serves as preliminary evidence of corporate data capitalization efforts. The measurement methodology subsequently: (1) aggregates all data-related capital expenditures within the fiscal year; (2) calculates their proportional representation relative to total fixed and intangible assets; and (3) derives a standardized ratio that quantifies the firm's investment in data asset development. This proportional metric functions as an empirical proxy for assessing the economic value of organizational data assets.

3.2.2. Independent variable

A common approach to measuring regional IP protection intensity is the ratio of patent infringement filings to case closures; however, this method primarily focuses on legal enforcement after an infringement occurs and does not adequately capture the proactive dimensions of IP protection. Another frequently used measure is the GP index, although its limited five-year time span may lack the necessary continuity and

could pose limitations in certain contexts (Ginarte & Park, 1997). From the perspective of market transaction theory, IP protection has a critical influence on fostering innovation by encouraging R&D investments and promoting high-quality outputs. It can also mitigate technological infringement and patent theft, ensuring a fair and competitive technology market. This study adopts the ratio of technology market transaction value to GDP as a comprehensive indicator of IP protection intensity, where a higher ratio signifies more effective protection, a more active technology market, and greater IP value relative to GDP.

3.2.3. Control variables

We also introduce a set of control variables to account for potential confounding factors, carefully operationalizing each variable based on established financial and governance metrics. Firm age is measured as the natural logarithm of years since a firm's establishment, and firm size is quantified using the natural logarithm of total assets. Financial structure is measured referencing leverage (total debt/total assets) and profitability by return on assets (net income/total assets). Asset composition is represented by tangible assets to total assets (physical asset intensity) and fixed assets to total assets (long-term investment ratio). Corporate governance controls include ownership concentration (largest shareholder's equity stake), executive alignment (CEO shareholding percentage), and board characteristics that include size (number of directors) and independence (proportion of independent directors). These variables, systematically presented in Table 2, are selected based on theoretical relevance to our research questions, established use in corporate finance literature, data availability, and measurement reliability, carefully considering potential multicollinearity issues.

3.3. Model construction

Our analytical approach uses a fixed effects (FEs) panel regression specification that includes temporal dynamics and cross-sectional heterogeneity in the sample. The empirical model is formally expressed as follows:

$$\text{Data capitalization}_{it} = \alpha + \beta \text{IPP}_{it} + \Sigma \text{control}_{it} + \Sigma \text{year} + \Sigma \text{industry} + \mu_i + \varepsilon_{it} \quad (1)$$

where $\text{Data capitalization}_{it}$ represents firm i 's degree of data capitalization in year t , which is the dependent variable. IPP_{it} denotes the IP protection level in the region where firm i is located in year t , and β is the regression coefficient. The model also includes firm-level control variables ($\Sigma \text{control}_{it}$) to control for additional factors that influence data breaches. Year FEs (Σyear) control for time-specific macroeconomic and

Table 1
Word spectrum of data asset identification.

Category	Specific Content
Data Collection	mobile internet, industrial internet, industrial Internet of Things (IoT), internet, internet+, internet technology, internet applications, internet business models, internet platforms, internet marketing, internet services, e-commerce, data services, online networks, networks, network marketing, online sales, network services, network technology, data management, information technology, information services, information networks, information-based management, information systems, information platforms, information sharing, information resources, information communication, information security, information management systems, informatization, communication technology, communication networks, mobile communication
	data mining, data models, data analysis, big data, text mining, intelligent data analysis, virtual reality, augmented reality, cloud computing, cloud services, cloud platforms, IoT, artificial intelligence, business intelligence, machine learning, deep learning, natural language processing, biometric technology, distributed computing
Data Storage	data storage, servers, data resources, data sharing, data management, databases, data warehouses, data centers, data chains, data services, data information, network resources, system integration, digitalization, digital platforms, digital networks, industrial intelligence, mobile intelligence, high-end intelligence, intelligent terminals, intelligent systems, intelligent technologies, intelligent hardware, intelligent management, smart homes, smart wearables, intelligent agriculture, intelligent investment advisors, intelligent cultural and tourism, intelligent customer service, intelligent technologies, intelligent transportation, intelligent buildings, intelligent manufacturing, intelligent robots, Internet-based medical care, intelligence, smartness, smart cities, mobile payments, digital marketing, digital control, digital communication
Data Collection	

Table 2
Variables definitions and measurement.

Variables	Definition
Data capitalization	Proportion of data assets in fixed and intangible assets disclosed in firms' annual reports
Intellectual property (IP) protection	Ratio of technology market turnover to GDP
Data capitalization resources	Number of patents related to digital technologies held by the firm
Data capitalization willingness	Frequency of keywords related to digital transformation in MD&A
Firm age	Ln (year since firms' foundation)
Firm size	Ln (total assets)
Leverage	Total debts/total assets
Return on assets	Net profit/total assets
Tangible assets	Tangible assets/total assets
Fixed assets	Tangible assets/total assets
Share concentration	Largest shareholders' shareholding ratio
CEO shareholding	CEOs' shareholding ratio
Board size	Number of board members
Board independence	Number of independent directors/Number of board members

regulatory factors that may affect all firms, and industry FEs ($\Sigma_{industry}$) capture industry-specific influences on data-breach risks. μ_i denotes individual FEs capturing individual, time-invariant characteristics, and ε_{it} is the random error term representing unaccounted random factors.

4. Results

4.1. Descriptive statistics

Table 3 presented the descriptive statistics. The data capitalization variable has a mean value of 0.329 (standard deviation (SD) = 0.070) with a range from 0 to 0.701, indicating that while digital assets represent about one-third of traditional asset holdings on average, significant variation exists across firms, encompassing from those yet to initiate digital transformation (minimum = 0) to early adopters leading the transition (maximum = 0.701). This distribution indicates that Chinese corporations are in a transitional phase, predominantly reliant on physical assets but gradually incorporating digital assets into their portfolios. Furthermore, the IP protection measure averages just 0.030 (SD = 0.043), with values spanning from 0 to 0.191, reflecting relatively underdeveloped technology markets overall with notable regional heterogeneity in innovation ecosystem maturity. These baseline patterns provide a crucial context for understanding the current landscape of corporate digital asset adoption and the institutional environment for knowledge commodification in China, establishing necessary benchmarks for subsequent causal analysis of the relationship between IP protection and data capitalization.

4.2. Benchmark regression

Table 4 presents the of benchmark regression results. The univariate specification (Model 1) establishes a statistically relevant positive association between IP protection and data capitalization ($\beta = 0.081$, $p < 0.01$). This fundamental relationship maintains its direction and significance ($\beta = 0.063$, $p < 0.01$) in the multivariate framework (Model 2), notwithstanding the 22.2 % coefficient reduction following incorporating organizational controls. While the coefficient decreases from 0.081 to 0.063, the core effect of IP protection on data capitalization remains robustly significant after introducing additional variables. These results support our theoretical framework, which posits that IP protection reduces uncertainties around data usage, increasing data assets' value and encouraging firms to accelerate data capitalization efforts. Overall, our baseline regression results strongly support H1.

4.3. Robustness tests

This study employs the generalized method of moments (GMM) to address the dual challenges of temporal persistence and endogeneity in estimating the intellectual property–data capitalization relationship.

Table 3
Descriptive statistics.

Variables	Obs	Mean	SD	Min	Max
Data capitalization	28,908	0.329	0.070	0	0.701
IP protection	28,908	0.030	0.043	0	0.191
Firm age	28,908	2.080	0.921	0	3.497
Firm size	28,908	22.320	1.503	14.942	31.175
Leverage	28,908	0.439	0.220	0.057	1.044
Return on assets	28,908	0.033	0.076	−0.341	0.275
Tangible assets	28,908	0.526	0.087	0.284	1
Fixed assets	28,908	0.191	0.151	0	0.937
Share concentration	28,908	0.333	0.150	0.003	0.900
CEO shareholding	28,908	0.057	0.120	0	0.800
Board size	28,908	8.523	1.774	0	21
Board independence	28,908	0.383	0.075	0.154	0.800
Data capitalization resources	18,640	1.693	1.455	0	9.449
Data capitalization willingness	28,722	1.554	1.243	0	6.380

Table 4
Baseline regression results.

Variables	(1)	(2)
IP protection	0.081*** (0.024)	0.063*** (0.024)
Firm age		−0.001 (0.001)
Firm size		0.008*** (0.001)
Leverage		−0.005*** (0.002)
Return on assets		−0.024*** (0.004)
Tangible assets		−0.045*** (0.004)
Share concentration		−0.015*** (0.003)
Fixed assets		−0.002 (0.004)
CEO shareholding		0.003 (0.004)
Board size		0.000* (0.000)
Board independence		0.004 (0.004)
Constant	0.348*** (0.015)	0.224*** (0.019)
N	28,908	28,908
R ²	0.203	0.221
Year FE	Included	Included
Industry FE	Included	Included

Note: * $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$. Standard errors are in parentheses.

This methodology incorporates several notable features, encompassing (1) lagged dependent variables to capture path-dependent accumulation effects in data asset development, (2) carefully constructed instrumental variables (IVs) to mitigate reverse causality concerns, and (3) a system GMM estimator that combines level and difference equations to improve efficiency. The results demonstrate that IP protection exhibits a strongly positive coefficient ($\beta = 0.192$, $p < 0.01$). Diagnostic tests confirm the validity of our IVs (Hansen J-test $p > 0.1$) and the absence of second-order autocorrelation (AR(2) $p > 0.1$), while Hansen tests validate the exogeneity assumptions of our system GMM specification. These findings validate our baseline results and provide stronger causal evidence confirming that IP protection promotes data capitalization. The methodological rigor of our approach substantially reduces biased estimation concerns and strengthens the policy relevance of our conclusions regarding the influence of IP protection institutional frameworks in fostering corporate data capitalization.

We use the random effects (RE) model to examine individual heterogeneity, assuming that individual heterogeneity is uncorrelated with the explanatory variables, which is a reasonable assumption in some panel data studies. By using the RE model, we can examine the stability of our research findings when treating individual differences as random disturbances. Compared with the baseline regression, the RE model is more efficient in using the time and individual dimensions of the panel data. Table 5 reveals that the coefficient for IP protection in the RE model is 0.102 ($p < 0.01$). Although the coefficient is smaller than in the GMM model, the direction of the effect remains consistent. The coefficients and significance of firm age and size are similar to those in the GMM model, further reinforcing our findings.

Replacing the dependent variable is a common method for testing research results' sensitivity to the measurement approach used. In practical research, different measurement methods may highlight different aspects of the research object. Substituting the dependent variable enables us to assess whether the research conclusions are dependent on a specific measurement approach. This study employs text mining to assign weights to the keyword frequencies listed in Table 1 using entropy weighting to calculate a subsequent composite index.

Table 5
Robustness tests.

Variables	(1) GMM	(2) Random	(3) DV	(4) IV
IP protection	0.192*** (0.056)	0.102*** (0.014)	0.410*** (0.115)	0.015** (0.007)
Firm age	−0.008*** (0.002)	−0.004*** (0.001)	−0.012*** (0.004)	−0.001 (0.001)
Firm size	0.010*** (0.002)	0.006*** (0.000)	0.013*** (0.002)	0.008*** (0.001)
Leverage	−0.002 (0.006)	−0.006*** (0.002)	−0.044*** (0.010)	−0.006*** (0.002)
Return on assets	0.002 (0.008)	−0.024*** (0.004)	−0.072*** (0.018)	−0.024*** (0.004)
Tangible assets	−0.037*** (0.012)	−0.051*** (0.004)	−0.126*** (0.020)	−0.045*** (0.004)
Share concentration	−0.027*** (0.010)	−0.026*** (0.003)	−0.085*** (0.015)	−0.016*** (0.003)
Fixed assets	−0.033*** (0.011)	−0.019*** (0.003)	−0.077*** (0.018)	−0.002 (0.004)
CEO shareholding	0.009 (0.009)	0.012*** (0.003)	−0.014 (0.018)	0.003 (0.004)
Board size	−0.001 (0.001)	0.000 (0.000)	−0.001 (0.001)	0.000* (0.000)
Board independence	0.011* (0.007)	0.005 (0.004)	0.018 (0.018)	0.004 (0.004)
L. Data assets	0.528*** (0.024)			
Constant	−0.027 (0.042)	0.277*** (0.014)	0.465*** (0.094)	0.210*** (0.020)
N	25,427	28,908	28,908	28,908
R ²	—	0.211	0.098	0.221
Year FE	Included	Included	Included	Included
Industry FE	Included	Included	Included	Included

Note: * $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$. Standard errors are in parentheses.

Table 5 reveals that the coefficient for IP protection increases

Table 6
Mechanism tests.

Variables	Data capitalization resources	Data capitalization	Data capitalization willingness	Data capitalization
IP protection	0.087* (0.049)	0.063** (0.030)	0.979** (0.474)	0.050** (0.023)
Data capitalization resources		0.027*** (0.005)		0.012*** (0.000)
Data capitalization willingness				−0.003*** (0.001)
Firm age	0.007*** (0.002)	0.003** (0.001)	0.127*** (0.016)	0.006*** (0.000)
Firm size	0.004*** (0.001)	0.006*** (0.001)	0.206*** (0.010)	−0.003* (0.002)
Leverage	−0.049*** (0.005)	−0.001 (0.003)	−0.155*** (0.042)	−0.022*** (0.004)
Return on assets	−0.022*** (0.007)	−0.011*** (0.004)	−0.185** (0.075)	−0.038*** (0.004)
Tangible assets	−0.006 (0.008)	−0.024*** (0.005)	−0.632*** (0.081)	−0.010*** (0.003)
Share concentration	−0.041*** (0.007)	−0.008* (0.005)	−0.538*** (0.063)	0.002 (0.004)
Fixed assets	−0.014 (0.009)	0.001 (0.006)	−0.317*** (0.074)	0.002 (0.004)
CEO shareholding	0.009 (0.007)	0.007 (0.004)	0.128* (0.074)	0.002 (0.004)
Board size	−0.001 (0.001)	0.000 (0.000)	0.031*** (0.005)	0.000 (0.000)
Board independence	0.003 (0.007)	−0.000 (0.004)	0.024 (0.072)	0.004 (0.004)
Constant	0.217*** (0.053)	0.191*** (0.033)	−3.935*** (0.386)	0.270*** (0.019)
Sobel test		3.011*** (0.104)		1.148** (0.547)
N	18,640	18,640	28,722	28,722
R ²	0.063	0.200	0.369	0.265
Year FE	Included	Included	Included	Included
Industry FE	Included	Included	Included	Included

Note: * $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$. Standard errors are in parentheses.

significantly to 0.410 ($p < 0.01$). Despite the significant change in the coefficient, the direction of the effect remains positive, indicating that our core conclusion remains robust across alternative dependent variable measures. This highlights the importance of carefully considering dependent variables' measurement method, which can significantly influence the research findings.

Using different measurement methods for the same independent variable enables us to examine whether the research results are biased by measurement errors or methodological choice. Our results confirm that IP protection fosters innovation by safeguarding expected returns. To capture this effect, regional IP protection in year $t - 1$ is measured by the ratio of IP lawsuits closed to lawsuits filed in each province during the previous year. As shown in Table 5, after replacement, the positive effect of IP protection on data capitalization remains significant, confirming that our research findings are robust regarding independent variable measurement.

5. Further analysis

5.1. Mechanism tests

To test H2 and H3, we conduct additional mechanism tests to explore whether IP protection promotes corporate data capitalization through firms' data capitalization resources and willingness.

We measure data capitalization resources using firms' number of patents related to digital technologies (Hsu & Ziedonis, 2013). These patents encompass areas such as data processing, cloud computing, and other innovations that directly contribute to enterprises' digital capabilities. Such patents reflect firms' investment in and commitment to digital technologies as an indicator of the capacity to leverage and capitalize on data-driven innovations. Examining the number of digital technology patents enables us to assess firms' ability to develop unique

digital assets, which is crucial for obtaining competitive advantages in the rapidly evolving digital landscape. Table 6 reveals a positive correlation between IP protection and resources investing in data capitalization ($\beta = 0.087$, $p < 0.1$). Next, we interact IP protection and data capitalization resources in the same equation to examine their relationship with data capitalization. The results reveal that data capitalization resources positively affect data capitalization ($\beta = 0.027$, $p < 0.01$), thus supporting H2. (See Table 6.)

We measure data capitalization willingness by analyzing the frequency of keywords related to digital transformation in firms' annual report management discussion and analysis (MD&A), applying a log-transformation to the keyword frequency measure after adding a unity constant to address the right-skewed distribution of raw word counts. This operationalization captures the presence and intensity of organizations' focus on digital initiatives. The empirical results in Table 6 demonstrate that a significantly positive correlation exists between IP protection intensity and strategic focus on digitalization ($\beta = 0.979$, $p < 0.05$), indicating that robust institutional environments enhance managerial attention to data capitalization. Furthermore, mediation analysis confirms that this strategic orientation positively influences actual data capitalization levels ($\beta = 0.050$, $p < 0.01$), validating the pathway proposed in H3.

5.2. Heterogeneity tests

We also investigate whether the above correlation varies along three key dimensions of firm characteristics, industrial sectors, and geographic regions.

Table 7 shows that IP protection positively affects private firms' data capitalization ($\beta = 0.079$, $p < 0.05$), whereas the relationship is insignificant for state-owned enterprises (SOEs). These differences reflect firms' distinct approaches to IP protection and data capitalization based on different ownership structures. Private firms are generally more agile in adapting business models and strategies, enabling them to quickly capitalize on the institutional environment and advance data

capitalization efforts. As a result, the relationship between IP protection and data capitalization are strongly positive and statistically significant for private firms. Conversely, with more rigid organizational structures and governance mechanisms, SOEs are often responsible for multiple strategic objectives that hinder their ability to pivot quickly toward data capitalization. Consequently, the influence of IP protection on data capitalization is less pronounced for SOEs.

Table 7 also shows that IP protection positively affects high-tech industries' data capitalization ($\beta = 0.100$, $p < 0.05$). In contrast, this relationship is insignificant for nonhigh-tech industries. These industry-specific differences are easily understandable as high-tech industries are fundamentally more reliant on technological innovation, particularly the adoption of cutting-edge technologies. In the digital economy, establishing competitive advantages through digital assets is crucial for high-tech firms. Such firms are more inclined to allocate resources toward developing and enhancing their data assets when protected by IP rights. Conversely, nonhigh-tech industries have less dependence on advanced data assets and digital technologies, typically focusing on incremental innovation to maintain market positions. As a result, such firms are less likely to leverage IP protection to drive investment in data capitalization.

Finally, Table 7 reveals that IP protection positively affects data capitalization ($\beta = 0.086$, $p < 0.05$) in regions with superior digital infrastructure. In contrast, the relationship is insignificant in regions with weaker digital infrastructure. These regional disparities are self-explanatory. In regions with well-developed digital infrastructure, firms have easier access to advanced digital technologies, facilitating rapid digital asset development. With the support of strong IP protection systems, firms in such regions are more likely to enhance their digital strategies, accelerating their data capitalization efforts. In contrast, firms in regions with underdeveloped digital infrastructure face significant barriers. Even with robust IP protection, the lack of necessary infrastructure limits their capacity and willingness to invest in data capitalization.

Table 7
Heterogeneity analysis regression results.

Variables	Ownership heterogeneity		Industry heterogeneity		Regional heterogeneity	
	State-owned	Private	High-tech	Nonhigh-tech	High foundation	Low foundation
IP protection	-0.034 (0.078)	0.079*** (0.026)	0.100*** (0.038)	0.035 (0.032)	0.086*** (0.031)	0.024 (0.042)
Firm age	0.003 (0.003)	-0.001 (0.001)	-0.002 (0.002)	0.001 (0.001)	-0.004*** (0.001)	0.002** (0.001)
Firm size	0.007*** (0.002)	0.009*** (0.001)	0.008*** (0.001)	0.008*** (0.001)	0.009*** (0.001)	0.007*** (0.001)
Leverage	-0.013 (0.011)	-0.006*** (0.002)	-0.004 (0.005)	-0.005** (0.002)	-0.004* (0.003)	-0.003 (0.004)
Return on assets	0.014 (0.020)	-0.027*** (0.004)	-0.003 (0.009)	-0.030*** (0.004)	-0.016*** (0.005)	-0.035*** (0.007)
Tangible assets	-0.082*** (0.021)	-0.040*** (0.004)	-0.065*** (0.011)	-0.039*** (0.004)	-0.053*** (0.005)	-0.038*** (0.007)
Share concentration	-0.020* (0.011)	-0.015*** (0.003)	-0.003 (0.006)	-0.014*** (0.004)	-0.018*** (0.004)	0.006 (0.006)
Fixed assets	0.046*** (0.014)	-0.005 (0.004)	0.010 (0.007)	-0.012** (0.005)	-0.012** (0.005)	0.014** (0.007)
CEO shareholding	0.013 (0.104)	0.003 (0.004)	0.065 (0.050)	0.004 (0.004)	0.013** (0.006)	-0.005 (0.005)
Board size	-0.000 (0.001)	0.001** (0.000)	-0.001 (0.000)	0.001*** (0.000)	0.000 (0.000)	0.001 (0.000)
Board independence	-0.015 (0.012)	0.004 (0.004)	0.001 (0.007)	0.004 (0.004)	0.007 (0.005)	-0.003 (0.005)
Constant	0.278*** (0.060)	0.207*** (0.020)	0.290*** (0.046)	0.205*** (0.026)	0.223*** (0.024)	0.237*** (0.029)
N	2625	26,283	8786	20,122	17,161	11,747
R ²	0.282	0.220	0.244	0.214	0.246	0.173
Year FE	Included	Included	Included	Included	Included	Included
Industry FE	Included	Included	Included	Included	Included	Included

Note: * $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$. Standard errors are in parentheses.

6. Discussion

We next discuss the empirical results. First, our empirical results support H1, confirming that IP protection promotes corporate data capitalization. Such protection establishes crucial institutional safeguards that reduce the risks associated with data leakage or theft while also encouraging innovation and digital resource aggregation. These findings are consistent with previous studies (Hu et al., 2022; Miric et al., 2019; Wen & Deng, 2023; Zheng et al., 2023) that have highlighted the influence of IP protection on fostering digital innovation, securing virtual assets, and ensuring firms' fair compensation. As IP protection strengthens, firms gain greater control over their data, establishing more competitive digital assets and driving the data capitalization process. This indicates that IP protection has a vital impact in safeguarding traditional technological innovations and advancing corporate data capitalization in the digital economy.

Second, the results reveal that IP protection enhances data capitalization through two distinct pathways of data capitalization resources (supporting H2) and data capitalization willingness (supporting H3). IP protection regulates data privacy and access, establishing a favorable institutional environment for firms (Hu et al., 2022). Under these protections, firms can effectively accumulate and use data resources through data mining, collection, analysis, storage, and sharing, which are essential for advancing data capitalization (Miric et al., 2019). Digital resource accumulation accelerates corporate investment in digital technologies, which is a critical factor in data capitalization. Additionally, IP protection reduces uncertainties in communication with external stakeholders, establishing a safer environment for information sharing (Miric et al., 2019) and supporting digital asset accumulation and capitalization. Corporate managers' willingness to invest in data capitalization is another significant pathway through which IP protection influences data capitalization. By mitigating uncertainties related to data usage rights, IP protection enhances firms' willingness to engage in digital innovation and transformation (Moerchel et al., 2022). When firms are confident in the safeguards provided by IP protection, this willingness is translated into active investment in digital technologies, further driving the data capitalization process (Hu et al., 2022). Moreover, IP protection reduces the risk of competitors' data theft or imitation, encouraging firms to allocate more resources toward digital assets (Lin, 2023; Sampat & Williams, 2019) rather than traditional physical assets. This signaling effect accelerates the pace of data capitalization.

Finally, heterogeneity tests reveal significant differences in the relationship between IP protection and data capitalization. First, IP protection has a positive effect on private firms' data capitalization, but not that of SOEs. Private firms are more agile and better positioned to leverage IP protection to increase their competitive edge or monopolistic advantages (Teece, 2018). In contrast, SOEs have more rigid governance structures and diverse strategic objectives and may struggle to expediently capitalize on these opportunities. Second, the relationship is significant for high-tech industries but not nonhigh-tech sectors. High-tech firms are more reliant on technological and knowledge-driven innovation (Zhang et al., 2024), and IP protection is crucial for their growth and digital transformation (Ghasemaghaei & Calic, 2019). Conversely, with less reliance on advanced data assets, nonhigh-tech industries do not prioritize IP protection in innovation strategies, making their impact on data capitalization less significant. Finally, IP protection has a significant effect on data capitalization in regions with well-developed digital infrastructure but not those with underdeveloped infrastructure. Firms can better leverage resources to enhance data capitalization in regions with advanced digital infrastructure, benefiting more from IP protection (Ravichandran et al., 2005). In contrast, firms in regions with underdeveloped infrastructure face barriers that hinder their ability to fully capitalize on IP protection.

6.1. Theoretical contributions

The empirical findings of this study demonstrate the positive impact of IP protection on corporate data capitalization, significantly contributing to the existing literature. First, the study extends the conceptual framework of IP protection by examining its influence in the context of digital assets. While previous research has distinguished between IP protection for tangible and intangible assets (Lin, 2023; Sampat & Williams, 2019), the specific mechanisms through which IP protection influences data capitalization have remained largely unexplored. Our results indicate that IP protection safeguards traditional technological innovations and has a crucial influence on the digital asset development and accumulation. This perspective enhances institutional theory, particularly in the digital economy era, by highlighting the role of IP protection in supporting digital asset development. It also deepens our understanding of how IP can facilitate data's transformation into valuable, monetizable assets, offering insights for scholars and practitioners navigating the complexities of data-driven business models.

Second, this study makes a valuable contribution to the literature on data capitalization. While previous research has widely acknowledged the central role of digital assets in establishing and sustaining competitive advantage (Hu et al., 2022; Kozjek et al., 2018; Lyytinen et al., 2016), relatively limited attention has been paid to the institutional factors that facilitate firms' effective use of these assets. This study expands upon previous work by identifying and examining accumulation and strategic use of data resources, and firms' willingness to invest in data-driven initiatives as two key mechanisms through which IP protection influences data capitalization. By confirming these mechanisms, we provide a deeper understanding of the organizational conditions and processes that enable firms to fully leverage digital assets. This not only advances our theoretical understanding of the factors driving data capitalization but also provides practical insights into how firms can effectively manage the complexities of data-driven innovation. Ultimately, this research enriches the broader discourse on digital innovation and data capitalization, highlighting the critical role of IP protection in shaping the ways in which firms manage and exploit digital assets for long-term competitive advantage.

6.2. Practical implications

The findings of this study yield several actionable recommendations for corporate leaders. Executives must not only acknowledge the strategic significance of digital technologies in the modern business environment but also learn to effectively leverage internal capabilities and external conditions—particularly the often-overlooked institutional environment—to support digital asset development. IP protection can mitigate risks such as data leakage, offering firms a valuable tool to enhance internal data governance, optimize systems for data management, and formulate strategies that unlock the full potential of digital resources. In addition, this study provides meaningful guidance for policymakers. Although IP protection has always had an important role, it has become even more vital in the era of digital transformation. Regulators and lawmakers involved in IP-related policymaking should prioritize establishing strongly enforced legal frameworks specifically designed to safeguard digital technologies. Considering the diversity across regional, industry, and firm characteristics, policy instruments should be differentiated, providing tailored incentives or regulatory support to strengthen IP protection where it is most needed. Such initiatives can establish a favorable institutional environment that promotes digital innovation to enable firms to use digital assets more effectively, contributing to the growth of the digital economy.

6.3. Limitations and future research

This study also opens promising directions for future inquiry. First, the analysis is based on a dataset of publicly listed firms in China, which

may constrain the generalizability of the findings to other national or regional contexts. Subsequent research could expand the scope to include firms from different countries, regions, or sectors to test the robustness and broader relevance of the relationships observed and exploring how IP protection influences data capitalization in various environments. Second, the effectiveness of IP protection differs significantly across jurisdictions. While some regions benefit from mature and sophisticated legal frameworks, others may still face institutional deficiencies. Future investigations could further investigate how institutional landscapes shape firms' ability to pursue data-driven innovation and asset development under different IP regimes. Such research would contribute to a more refined understanding of how contextual legal factors moderate the impact of IP protection on digital transformation. Finally, although this study highlights data resource accumulation and firms' willingness to invest in data-related initiatives, additional mechanisms through which IP protection enhances corporate data capitalization may be at play. Further research could identify other pathways and examine how they differs across industries or regions, offering a more comprehensive understanding of how IP protection supports digital asset development across diverse settings.

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Data availability

Data will be made available on request.

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